PARTIAL CIRCUIT MILESTONE — NOT A RELEASE DESIGN

Global nets join sheets. GEN_ and HUB_ segregate control and phase nets. DC_LINK / PGND shared.

36–58.4 V screening range only. No validated current rating. No PCB or routing.

Protected entry still missing: OV chopper and fault memory populated on dedicated pages.

GEN DRV8353S driver and three low-side current channels

File: gen-driver.kicad_sch

GEN 100 V MOSFET bridge — schematic only

File: gen-bridge.kicad_sch

HUB DRV8353S driver and three low-side current channels

File: hub-driver.kicad_sch

HUB 100 V MOSFET bridge — schematic only

File: hub-bridge.kicad_sch

Unfinished safety / control / supply boundaries

File: external-boundaries.kicad_sch

Bus-fed auxiliary and independent safety supply

File: aux-supply.kicad_sch

Autonomous OV chopper comparator and rail / brake monitors

File: protection-comparators.kicad_sch

Set-dominant fault memory and hardware bridge enable

File: fault-latch.kicad_sch

Independent brake chopper power stage

File: brake-chopper.kicad_sch

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /
File: powerboard-reva.kicad_sch

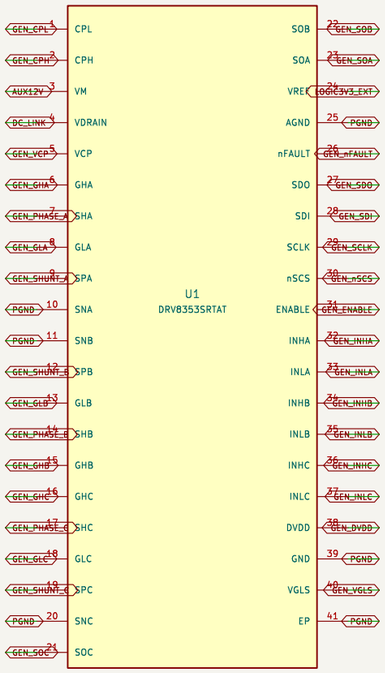
Title: Pedal-by-wire: hierarchical engineering review

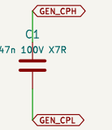
Size: A2	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 1/10

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

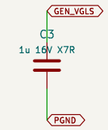
VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

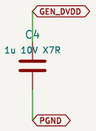
RELEASE BLOCKERS: MCU, fast OC protection and transient/thermal qualification unfinished.

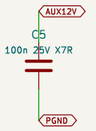


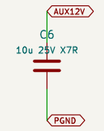






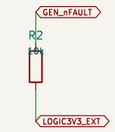


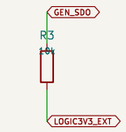


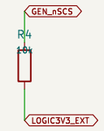












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

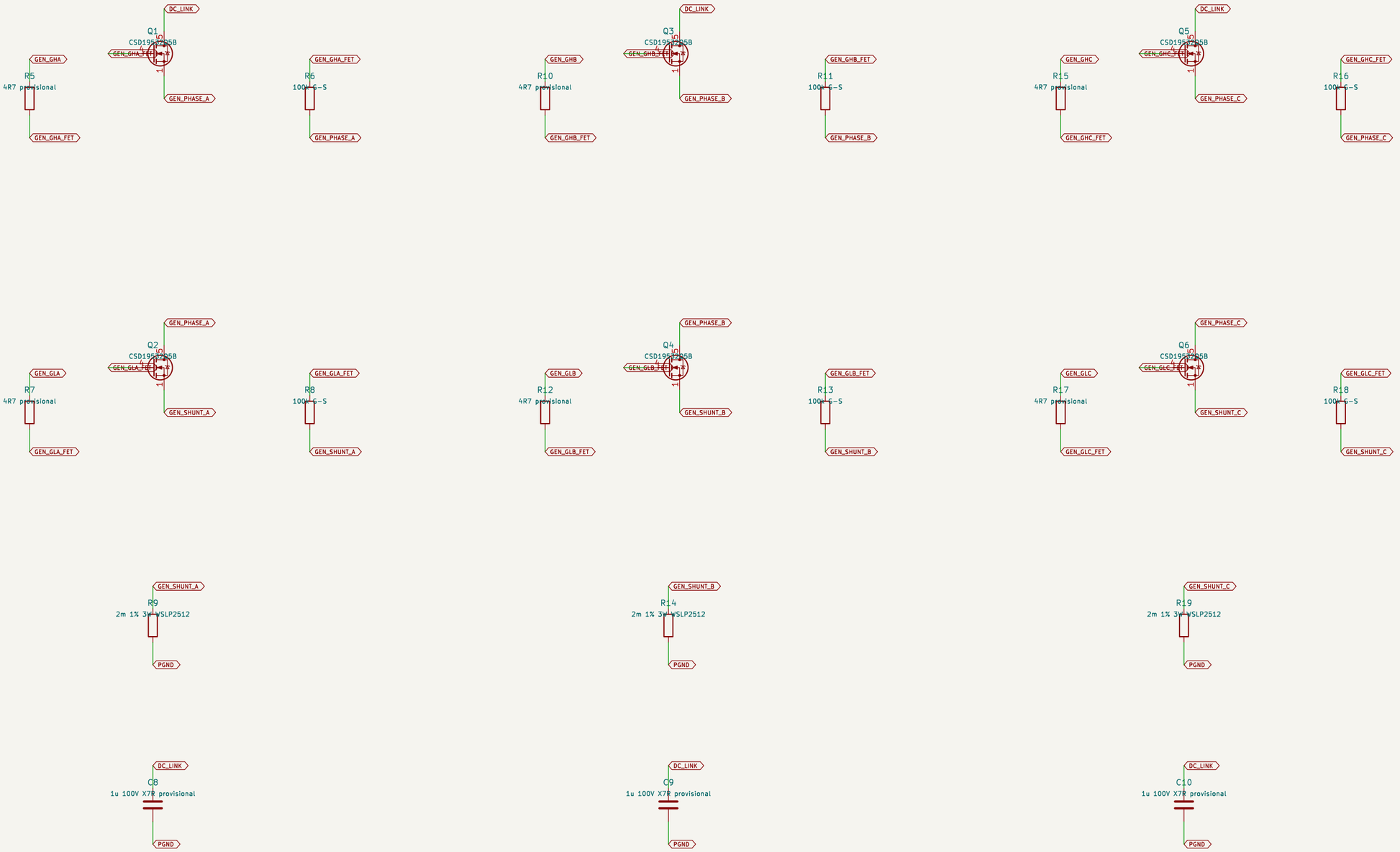
Sheet: /GEN DRV8353S driver and three low-side current channels/
File: gen-driver.kicad_sch

Title: GEN DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 2/10

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /GEN 100 V MOSFET bridge — schematic only/
File: gen-bridge.kicad_sch

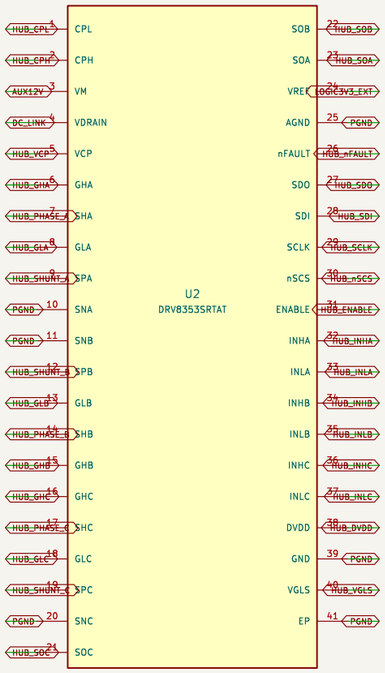
Title: GEN 100 V MOSFET bridge — schematic only

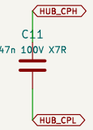
Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 3/10

TI SLVSDY6A pp6–7 pin table. 40-pin RTA + exposed pad. SPI variant, NOT hardware variant.

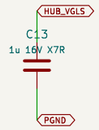
VM supplied by bus-fed auxiliary 12 V. VREF=3V3 (CSA midpoint internally VREF/2).

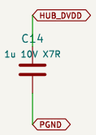
RELEASE BLOCKERS: MCU, fast OC protection and transient/thermal qualification unfinished.







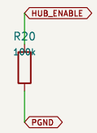


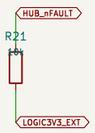


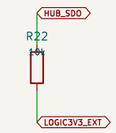


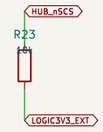












ENABLE pull-down defaults off; not a safety-rated inhibit. SPI configuration and CSA gain20 not yet validated.

NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

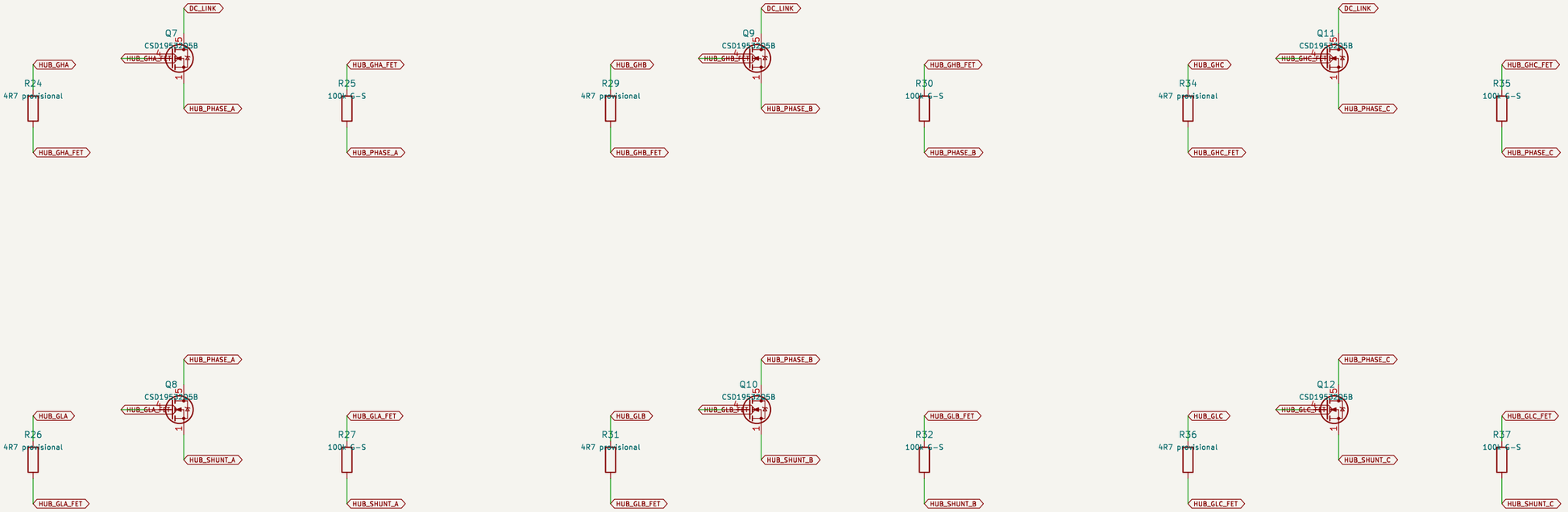
Sheet: /HUB DRV8353S driver and three low-side current channels/
File: hub-driver.kicad_sch

Title: HUB DRV8353S driver and three low-side current channels

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 4/10

CSD19532Q5B stock KiCad symbol / pad map. Gates: 4; drains: 5; sources: 1,2,3.

Labels are electrical connections. Rgate=4R7 provisional. Kelvin shunts: no routing qualified.



Shunt amplifier SPx taps SHUNT_x; SNx taps PGND: Kelvin layout mandatory. 15A RMS => 0.45W per shunt.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /HUB 100 V MOSFET bridge — schematic only/
File: hub-bridge.kicad_sch

Title: HUB 100 V MOSFET bridge — schematic only

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 5/10

OFFBOARD interfaces: MCU signals, brake loop and protected bus entry.

DO NOT ENERGIZE. Protected entry and MCU unfinished; populated protection remains unqualified.

J1
PROTECTED_DC_INPUT_EXT
DC_LINK
PGND

J3
GEN_PHASE_A
GEN_PHASE_B
GEN_PHASE_C
_PHASES

J7
HUB_PHASE_A
HUB_PHASE_B
HUB_PHASE_C
_PHASES

J2
MCU_SUPPLY_EXT
LOGIC3V3_EXT
PGND

J4
GEN_INHA
GEN_INLA
GEN_INHB
GEN_INLB
GEN_INHC
GEN_INLC
PGND
_PWM_EXT

J8
HUB_INHA
HUB_INLA
HUB_INHB
HUB_INLB
HUB_INHC
HUB_INLC
PGND
_PWM_EXT

J5
RESET_MCU_OPEN
MCU_ARM
WDI_SAFETY
TIMER_BREAK_CLK
SAFETY3V3
BRAKE_LOOP
PGND
_CONTROL

J9
GEN_nSCS
GEN_SCLK
GEN_SDI
GEN_SDO
GEN_nFAULT
PGND
_SPI_EXT

J11
HUB_nSCS
HUB_SCLK
HUB_SDI
HUB_SDO
HUB_nFAULT
PGND
_SPI_EXT

DC_LINK
C21
100u 100V provisional
PGND

J6
GEN_SOA
GEN_SOB
GEN_SOC
PGND
_ADC_EXT

J10
HUB_SOA
HUB_SOB
HUB_SOC
PGND
_ADC_EXT

Missing: source fuse, reverse blocking, precharge, DC contactor, bus current sensor and discharge.

Populated: OV chopper, fault latch, brake loop, watchdog, aux supply. Missing: fast current trip.

Missing: STM32 timer/ADC mux, Hall/encoder sensors, thermal sensors, storage/BMS charge acceptance.

PGND is common, not galvanically isolated. Supply sequencing and back-power paths unqualified.

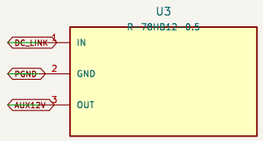
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

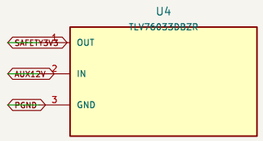
Sheet: /Unfinished safety / control / supply boundaries/
File: external-boundaries.kicad_sch

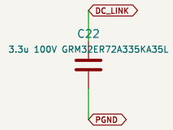
Title: Unfinished safety / control / supply boundaries

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 6/10

R-78HB12-0.5: 17–72V operating, 75V absolute. Entry transient protection still REQUIRED.



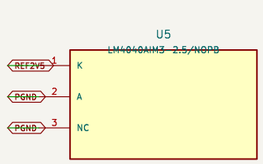




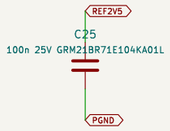












Safety supply is NOT the MCU supply. LOGIC3V3_EXT remains a supervised, external MCU rail.

Supply total/current/thermal budget provisional; no galvanic isolation or stored-energy ride-through.

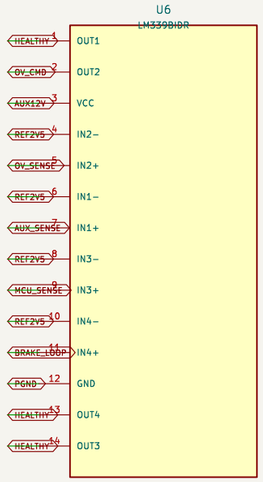
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

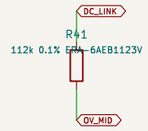
Sheet: /Bus-fed auxiliary and independent safety supply/
File: aux-supply.kicad_sch

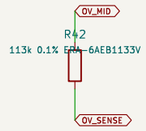
Title: Bus-fed auxiliary and independent safety supply

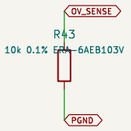
Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 7/10

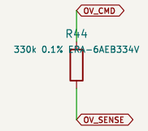
Comparator powered from AUX12V; open collectors pulled to SAFETY3V3. No MCU in OV path.



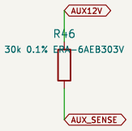


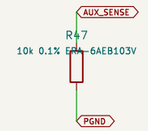


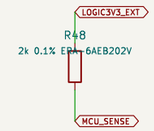


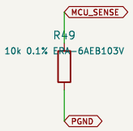


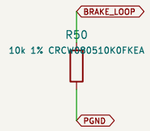




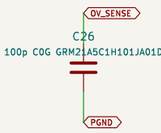


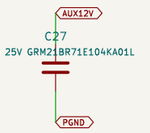












BRAKE_LOOP must close to SAFETY3V3 through normally-closed brake + thermal switches. Open wire trips.

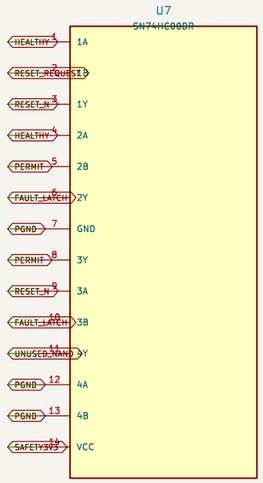
NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

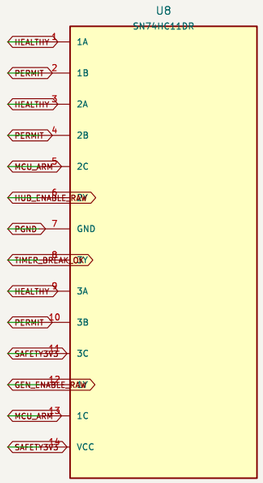
Sheet: /Autonomous OV chopper comparator and rail / brake monitors/
File: protection-comparators.kicad_sch

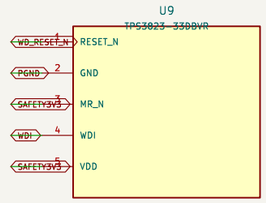
Title: Autonomous OV chopper comparator and rail / brake monitors

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 8/10

HEALTHY low SETS fault; RESET_REQUEST accepted only while HEALTHY high. ARM alone cannot reset.



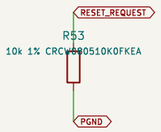


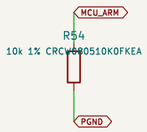




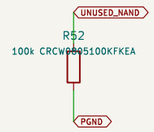


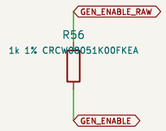


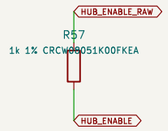


















Watchdog WDI cannot float: fitted 10k pulldown. 0.9–2.5s timeout is NOT short-circuit protection.

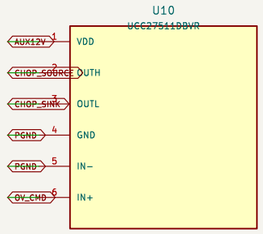
NOT SAFE TO ENERGIZE – SAFETY CIRCUITS UNFINISHED

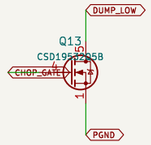
Sheet: /Set-dominant fault memory and hardware bridge enable/
File: fault-latch.kicad_sch

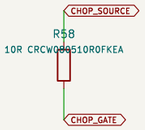
Title: Set-dominant fault memory and hardware bridge enable

Size: A3	Date:	Rev: A-PARTIAL
KiCad E.D.A. 9.0.9		Id: 9/10

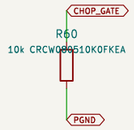
Autonomous OV path ignores bridge ENABLE. External resistor is physically required across J12 pins.

















Candidate: 2 x TE HSC3004R7J in series (9.4 ohm). Chassis/heatsink + NC thermal cutoff required.

No resistor cable clamp / fuse / SOA qualification yet. Loss of AUX supply loses bus sink: NOT FAIL SAFE.

NOT SAFE TO ENERGIZE — SAFETY CIRCUITS UNFINISHED

Sheet: /Independent brake chopper power stage/
File: brake-chopper.kicad_sch

Title: Independent brake chopper power stage

Size: A3	Date:	**Rev: A-PARTIAL**
KiCad E.D.A. 9.0.9		Id: 10/10